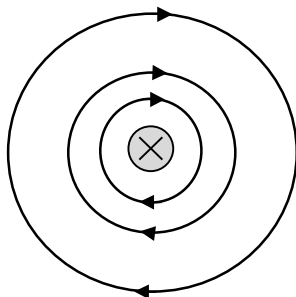


energy of the reactants (the two smaller nuclei).

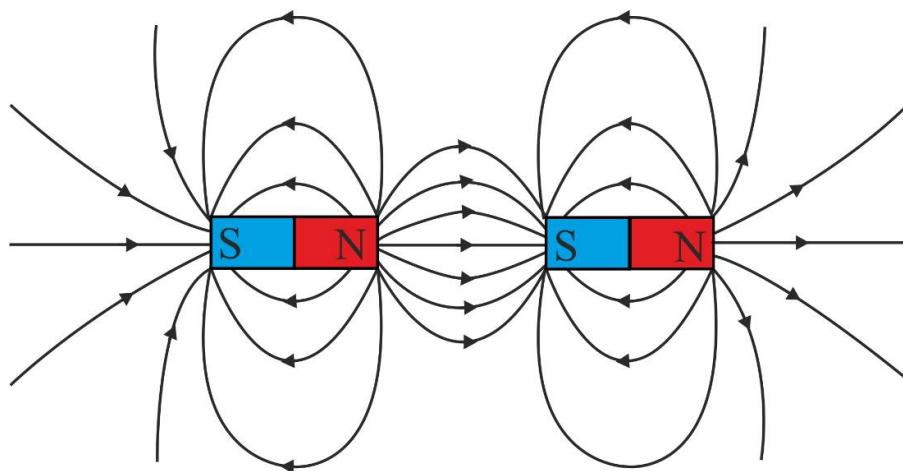
6(a)

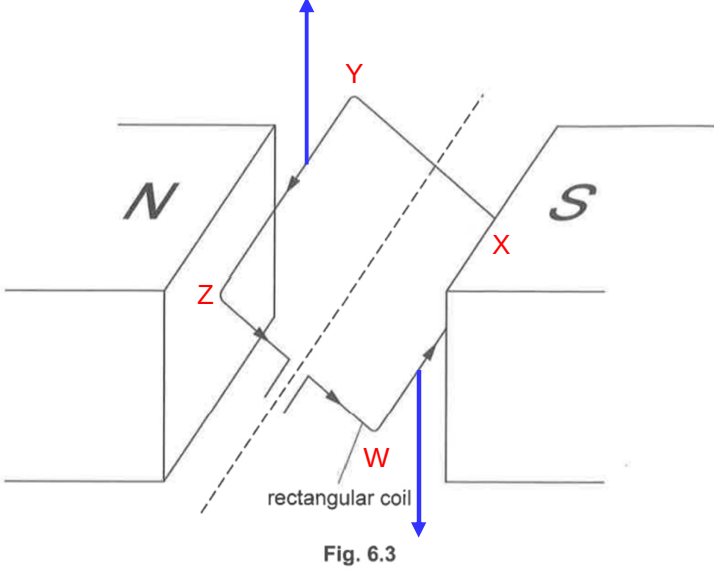
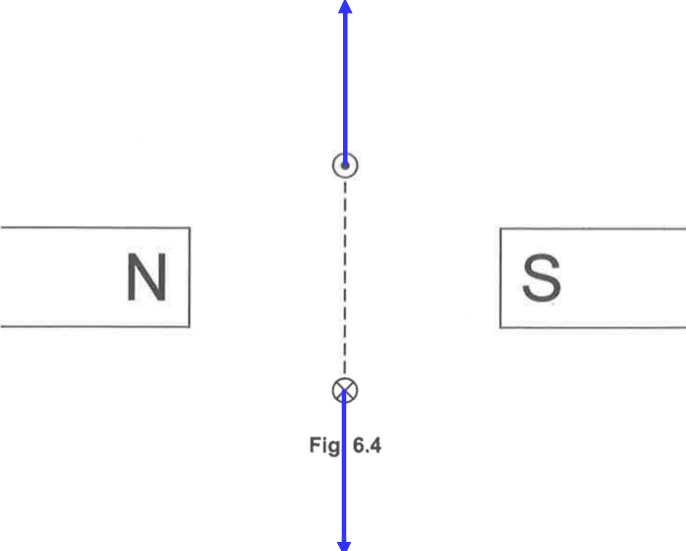
Magnetic flux density is defined as the force per unit length per unit current acting on a straight conductor placed perpendicularly to the magnetic field.

6(b)(i)



6(b)(ii)



No.	Solution
6(c)(i)	 Fig. 6.3
6(c)(ii)	Assuming that the coil is in a horizontal position, torque of the couple $= (0.85)(1.25)(0.040) \sin 90^\circ (0.040)$ $= 1.7 \times 10^{-3} \text{ N m}$
6(d)(i)	 Fig 6.4
6(d)(ii)	As the coil rotates from the position shown in Fig. 6.3 to the position shown in Fig. 6.4, the perpendicular distance between the lines of action of the magnetic force on WX and the magnetic force on YZ decreases to zero. Although the magnetic force is unchanged, the torque on the coil decreases to zero when in the position shown in Fig. 6.4.

No.	Solution
6(e)(i)	Resistance of coil $= \frac{5.6}{1.25} = 4.5 \text{ } \Omega$
6(e)(ii)	$R = \frac{\rho NL}{A}$ $N = \frac{RA}{\rho L} = \frac{(4.48) \pi (0.05 \times 10^{-3})^2}{(1.68 \times 10^{-8})(4 \times 0.040)} = 13.09 \approx 13 \text{ turns}$
7(a)	Electric field strength at a point is defined as the electric force exerted per unit